Guleed A. Ali

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RESEARCH PROGRAM

Quaternary geology, isotope geochemistry, and geochronology applied to abrupt hydroclimate change, ice-sheet and glacier chronologies, and Earth-surface processes. Active field programs in the United States are in the Great Basin, Grand Canyon, Olympic Peninsula, and Long Island.

Co-PI of Stony Brook's isotope geochemistry facility (FIRST Lab): Nu Plasma 3 MC-ICP-MS, Agilent 8900 triple-quadrupole ICP-MS, IsotopX Phoenix TIMS. Select methods: U-Th and U-Pb geochronology, boron isotopes ($\delta^{11}B$).

APPOINTMENTS

2025–	**Assistant Professor**, Geosciences, Stony Brook University (SUNY)
2023–25	**IDEA Fellow**, Stony Brook University
2022–23	**Postdoctoral Researcher**, Berkeley Geochronology Center and UC Berkeley with Warren Sharp
2019–21	**Research Fellow**, Earth Observatory of Singapore, Nanyang Technological University with Xianfeng Wang and Kerry Sieh
2018–19	**Ford Foundation Postdoctoral Fellow**, University of Maine with Aaron Putnam

EDUCATION

2018	**Ph.D.**, Earth and Environmental Sciences, Columbia University *Late Glacial and Deglacial Fluctuations of Mono Lake, California* · advisor Sidney Hemming
2010	**B.S.**, Geology, *cum laude*, University of Arizona · advisor Peter Reiners

RESEARCH SUPPORT

2026–	**U.S. Geological Survey** · Surficial geologic mapping of the Montauk quadrangle, Long Island, NY · PI
2026–	**Washington State** · Cordilleran Ice Sheet chronology and surficial mapping, northwest Washington · PI

PEER-REVIEWED PUBLICATIONS

in prep.	**Ali, G.A.**, Lin, K., Hemming, S.R., Putnam, A.E., Sharp, W., Wang, X. Heinrich Event 1 caused the wettest century in the western US in at least 70,000 years.
2026	**Ali, G.A.**, Lin, K., Hemming, S.R., Wang, X. Carbonate U/Th ages confirm the Mono Basin excursion as the Laschamps geomagnetic excursion. *GSA Bulletin* 138, 2910–2918. doi:10.1130/B38358.1
2022	**Ali, G.A.H.**, Lin, K., Hemming, S.R., Cox, S.E., Ruprecht, P., Zimmerman, S.R.H., Stine, S., Wang, X. Emergence of wet conditions in the Mono Basin of the western USA coincident with inception of the Last Glaciation. *GSA Bulletin* 134, 2267–2279. doi:10.1130/B36084.1
2020	Hodelka, B.N., McGlue, M.M., Zimmerman, S., **Ali, G.**, Tunno, I. Paleoproduction and environmental change at Mono Lake (eastern Sierra Nevada) during the Pleistocene–Holocene transition. *Palaeogeogr. Palaeoclimatol. Palaeoecol.* 543, 109565. doi:10.1016/j.palaeo.2019.109565
2019	Hudson, A.M., Hatchett, B.J., Quade, J., Boyle, D.P., Bassett, S.D., **Ali, G.**, De los Santos, M.G. North–south dipole in winter hydroclimate in the western United States during the last deglaciation. *Scientific Reports* 9, 4826. doi:10.1038/s41598-019-41197-y

2017 Fu, R.R., Weiss, B.P., Lima, E.A., Kehayias, P., Araujo, J.F., Glenn, D.R., Gelb, J., Einsle, J.F., Bauer, A.M., Harrison, R.J., **Ali, G.A.** Evaluating the paleomagnetic potential of single zircon crystals using the Bishop Tuff. *Earth Planet. Sci. Lett.* 458, 1–13. doi:10.1016/j.epsl.2016.09.038

2017 Hudson, A.M., Quade, J., **Ali, G.**, Boyle, D., Bassett, S., Huntington, K.W., De los Santos, M.G., Cohen, A.S., Lin, K., Wang, X. Stable C, O and clumped isotope systematics and 14C geochronology of carbonates from the Quaternary Chewaucan closed-basin lake system, Great Basin, USA. *Geochim. Cosmochim. Acta* 212, 274–302. doi:10.1016/j.gca.2017.06.024

2009 **Ali, G.A.H.**, Reiners, P.W., Ducea, M.N. Unroofing of Alabama and Poverty Hills basement blocks, Owens Valley, California from apatite (U-Th)/He thermochronology. *Int. Geol. Rev.* 51, 1034–1050. doi:10.1080/00206810902965270

CONFERENCE PRESENTATIONS, 2024–PRESENT

Presenting author first. * student presenter.

GSA 2026 *Elbaroudy, L., **Ali, G.A.** Reconciling radiocarbon and cosmogenic chronologies of the Laurentide Ice Sheet retreat across Long Island, New York. Poster 55-4, GSA Connects, Denver, 11–14 Oct.

AGU 2025 **Ali, G.A.**, Creel, R., McKenzie, M., Nguyen, H. Revisiting the last Juan de Fuca Lobe maximum using radiocarbon dating of sub-fossil logs across the Olympic Peninsula, northwest Washington. PP41E-1200, New Orleans, 15–19 Dec.

AGU 2024 Munoz-Saez, C., **Ali, G.A.**, Lin, K., Wang, X. U/Th ages of travertine deposition in the eastern Sierra Nevada. PP33C-0564, Washington, DC, 9–13 Dec.

INVITED TALKS

Sep 2026 Lamont-Doherty Earth Observatory & Columbia University

Sep 2025 **Pegrum Lecture**, Department of Earth Sciences, University at Buffalo

May 2025 Department of Geosciences, University of Arizona

Apr 2025 Department of Earth Science and Environmental Change, University of Illinois Urbana-Champaign

Apr 2025 **Smith Lecture**, Department of Earth and Environmental Sciences, University of Michigan

ADVISING AND MENTORING

2026– **Zahra Omarjee**, Ph.D. student · M.S., University of KwaZulu-Natal, South Africa

2025– **Laila Elbaroudy**, undergraduate researcher · deglacial chronology of the Laurentide Ice Sheet, Long Island
GSA Northeastern Section Undergraduate Research Grant, 2026 · poster, GSA Connects 2026

TEACHING

Fall 2026 **Introduction to Physical Geology**, instructor, Stony Brook University

Spring 2026 **Introduction to Physical Geology**, instructor, Stony Brook University

Fall 2025 **Architecture of Time**, instructor, Stony Brook University

Fall 2024 **Introduction to Physical Geology**, instructor, Stony Brook University

Fall 2023 **Introduction to Physical Geology**, co-instructor, Stony Brook University

Spring 2022 **Excursion to the Eastern Sierra Nevada**, instructor, Columbia University and Barnard College

2018, 2019 Summer Field Camp, teaching assistant, Earth Observatory of Singapore / NTU

2011–24 Outdoor Education Program, Mono Lake Committee, geology lecturer and outdoor guide

SERVICE

2023– EDI Committee, Department of Geosciences, Stony Brook University

2015 Co-convener, *Observations and Dynamics of Deglaciations*, AGU Fall Meeting

2013	Co-convener, *Great Basin Hydroclimate*, GSA Annual Meeting

AWARDS AND HONORS

2018–19	Ford Foundation Postdoctoral Fellowship
2015	Ford Foundation Dissertation Fellowship, honorable mention · GSA Graduate Student Research Grant
2012–15	NSF Graduate Research Fellowship
	2012 NSF East Asia and Pacific Summer Institute (declined)
2010	Patricia Hall Award, National Association of Black Geologists and Geophysicists
2009	Outstanding Senior Award, Geosciences, University of Arizona · Best Student Paper, NABGG · Galileo Circle Scholar
2008–10	Ronald E. McNair Scholar
2008	Len Assante Scholarship, National Groundwater Association · Excellent Academics Award, NABGG · Minority Participation Program, American Geosciences Institute